INCH-POUND

MIL-PRF-32711
w/ AMENDMENT 1
14 February 2024
SUPERSEDING
MIL-PRF-32711
14 June 2022

PERFORMANCE SPECIFICATION

DIAMOND-LIKE CARBON (DLC) COATING FOR ARMAMENT WEAPONS SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense within the limitations noted at the bottom of the page.

1. SCOPE

1.1 Scope. This specification establishes the requirements for an amorphous, solid, Diamond-Like Carbon (DLC) coating bonded to ferrous metal Armament Weapon Systems for the purpose of reducing wear, prevent galling, corrosion, and seizure of metals (see 6.1). The DLC coating is referred to as coating herein.

1.2 Classification. The DLC coatings are of the following classes and grades, as specified.

1.2.1 Classes.

Class I: Thickness not less than 2.5 µm to less than 5.0 µm.
Class II: Thickness not less than 5.0 µm to less than 10.0 µm.
Class III: Thickness not less than 10.0 µm to not greater than 20.0 µm.

1.2.2 Grades.

Grade A: not less than 8 GPA to less than 12 GPA.
Grade B: not less than 12 GPA to not greater than 20 GPA.

2. APPLICABLE DOCUMENTS

Comments, suggestions or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, NJ 07806-5000, or emailed to: usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of information using the ASSIST database at https://assist.dla.mil.

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2.1 General The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL SPECIFICATIONS

VV-D-1078B	-	Damping Fluid, Silicone Base (Dimethyl Polysiloxane)

FEDERAL STANDARDS

FED-STD-791	-	Testing Method of Lubricants, Liquid Fuels, And Related Products

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-PRF-372	-	Cleaning Compound, Solvent (For Bore Of Small Arms And Automatic Aircraft Weapons)
MIL-PRF-14107	-	Lubricating Oil, Weapons, Low Temperature
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Numbers: O-152, O-154, O-156, and O-167
MIL-PRF-46170	-	Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base, NATO Code No. H-544
MIL-PRF-63460	-	Cleaner, Lubricant, and Preservative for Weapons and Weapons Systems

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-171	-	Finishing of Metal And Wood Surfaces

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

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US ARMY Combat Capabilities Development Command Armament Center

13104637	-	Conformity Coupon

(Copies of these drawings may be requested from DEVCOM Armaments Center FCDD-ACE-SP by email at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM A108-18	-	Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A322-13	-	Standard Specification for Steel Bars, Alloy, Standard Grades
ASTM B117-19	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B499	-	Standard Test Method for Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals
ASTM B571-18	-	Standard Practice for Qualitative Adhesion Testing of Metallic Coatings
ASTM D1141-98	-	Standard Practice for the Preparation of Substitute Ocean Water
ASTM D1193-06	-	Standard Specification for Reagent Water
ASTM D2510-94	-	Standard Test Method for Adhesion of Solid Film Lubricants
ASTM D2511-20	-	Standard Test Method for Thermal Shock Sensitivity of Solid Film Lubricants
ASTM D7091-21	-	Standard Practice for Nondestructive Measurement of Dry Film Thickness of Nonmagnetic Coatings Applied to Ferrous Metals and Nonmagnetic, Nonconductive Coatings Applied to Non-Ferrous Metals
ASTM E3-11	-	Standard Guide for Preparation of Metallographic Specimens.
ASTM E384-17	-	Standard Test Method for Microindentation Hardness of Materials
ASTM E2546-15	-	Standard Practice for Instrumented Indentation Testing
ASTM F22-21	-	Standard Test Method for Hydrophobic Surface Films by the Water-Break Test
ASTM G99-17	-	Standard Test Method for Wear Testing with a Pin-on-Disk Apparatus

(Copies of these documents are available from https://astm.org.)

SAE INTERNATIONAL

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AMS 1424/1	-	Deicing/Anti-Icing Fluid, Aircraft SAE Type I Glycol (Conventional and Non-Conventional) Based
AMS-QQ-A-367	-	Aluminum Alloy Forgings

(Copies of these documents are available from https://sae.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections

3.1.1 Process prove out. When specified (See 6.2), samples shall be subjected to Process prove out in with accordance 4.2.

3.1.2 First article inspection. When specified (See 6.2), samples shall be subjected to First article inspection in with accordance 4.3.

3.1.3 Conformance inspection. Samples shall be subjected to conformance inspection in accordance with 4.4.

3.2 Support and ownership requirements.

3.2.1 Part coating thickness. The part's average coating thickness shall be in accordance with the Class listed in the applicable drawing (see 1.2.1). This test is only performed during first article inspection and conformance inspection on the part in the applicable drawing.

3.2.2 Conformity coupon coating thickness.

a. For the conformity coupon (DWG 13104637), no individual coating thickness reading shall exceed ±25% of the average thickness for Class III coatings (see 1.2.1). This test is only performed during process prove out on conformity coupons (see 4.5.2).

b. For the conformity coupon (DWG 13104637), no individual coating thickness reading shall exceed ±50% of the average thickness for Class I and II coatings (see 1.2.1). This test is only performed during process prove out on conformity coupons (see 4.5.2).

3.2.3 Coating adhesion. The coating adhesion shall be compliant with ASTM B571-18, modified scribe test (see 4.6.3). For process prove out and first article inspection, test panels (see 4.5.2) shall be used. For first article and conformance inspection, the part in the applicable drawing shall be used.

3.2.4 Coating hardness. The coating hardness shall be in accordance with the Grade listed in the applicable drawing (see 1.2.2). For process prove put, test panels (see 4.5.2) shall be used. For first article inspection and conformance inspection, the part in the applicable drawing shall be

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used.

3.2.5 Coefficient of friction. This test is only performed during process prove out on pins and discs/plates (see 4.5.2).

a. The coating when applied to 8620 carburized steel panel (see 4.5.2) shall have a kinetic coefficient of friction less than or equal to 0.2 when dry sliding against UNCOATED 8620 carburized steel pin. The coefficient of friction shall be maintained in relative humidity ranges of 10-90%.

b. The coating when applied to 8620 carburized steel panel (see 4.5.2) shall have a kinetic coefficient of friction less than or equal to 0.2 when dry sliding against COATED 8620 carburized steel pin (see 4.5.2). The coefficient of friction shall be maintained in relative humidity ranges of 10-90%.

c. The coating when applied to 8620 carburized steel panel (see 4.5.2) shall have a kinetic coefficient of friction less than or equal to 0.2 when dry sliding against anodized 7075 aluminum pin (as per AMS-QQ-A-367 condition T6 with Protective finish 7.5.2 of MIL-STD-171). The coefficient of friction shall be maintained in relative humidity ranges of 10-90%.

3.2.6 Resistance to fluids. After immersion in each of the test fluids specified in Table I, the coating shall not degrade to the point that bare metal of the coated item is exposed. The coating shall meet hardness (see 3.2.4) after being immersed in the test fluids. In addition, the coating shall not lift, blister, crack, or peel after being immersed in the test fluids. This test is only performed during process prove out on test panels (see 4.5.2).

TABLE I. Test fluids.

Test Fluid	Specification
Anti-Icing Fluid	AMS 1424/1
Cleaning Compound, Solvent for Bore of Small Arms and Automatic Weapons	MIL-PRF-372
Reagent Water	ASTM D1193-06, Type III
Substitute Ocean Water	ASTM D1141-98
Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base	MIL-PRF-46170
Turbine Fuel, Aviation, Kerosene Types	MIL-DTL-83133, JP-8
Lubricating Oil, Aircraft Turbine Engine, Synthetic Base	MIL-PRF-23699
Lubricating Oil, Weapons, Low Temperature	MIL-PRF-14107
Lubricant, Cleaner and Preservative for Weapons and Weapon Systems	MIL-PRF-63460
Damping Fluid, Silicone Base (Dimethylpolysiloxane)	VV-D-1078

3.2.7 Workmanship. The coating shall appear uniform in color, smooth, free from any cracks, scratches, pinholes, blisters, foreign matter, grit, or other particulate. For process prove out, test panels (see 4.5.2) and conformity coupons (see 4.5.2) shall be examined. For first article and

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conformance inspection, the part in the applicable drawing shall be examined.

3.3 Environmental requirements.

3.3.1 Maximum operating temperature. When exposed to temperatures up to 350° C for 60 minutes in air, the sample, mounted on a wire rack, shall conform to the coefficient of friction (3.2.5) and adhesion requirements (see 3.2.3). This test is only performed during process prove out on test panels (see 4.5.2).

3.3.2 Thermal stability. The coating's thermal stability shall be compliant with ASTM D2511-20. There shall be no evidence of cracks, scratches, pinholes, blisters, foreign matter, grit, or other particles. This test is only performed during process prove out on test panels (see 4.5.2).

3.3.3 Humidity resistance. The coating shall protect the panels so that the total number of corrosion dots on all three panels (combined) is not greater than three after 900 hours exposure in a humidity cabinet. No individual corrosion dot shall exceed one millimeter in length, width, or diameter. This test is only performed during process prove out on test panels (see 4.5.2).

3.3.4 Salt-spray (5 % solution) resistance. The coating shall protect the panels so that the total number of corrosion dots on each test panel is not greater than three after 100 hours exposure to a spray of 5 % salt solution. No individual corrosion dot shall exceed one millimeter in length, width, or diameter. This test is only performed during process prove out on test panels (see 4.5.2).

3.3.5 Water displacement and water stability. The coating shall displace water so that there is no evidence of rust spots, staining or pitting due to corrosion on the test panels after storage in the static humidity chamber for one hour. This test is only performed during process prove out on test panels (see 4.5.2).

3.4 Operating requirements.

TABLE II. Weapon system operating requirement.

Weapon System	Verification
CARBINE, 5.56 MILLIMETER: M4A1	See Appendix A

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4. VERIFICATION

TABLE III. Requirement/verification cross reference matrix.

METHOD OF VERIFICATION 1 - Analysis 2 - Demonstration 3 - Examination 4 - Test							CLASSES OF VERIFICATION A - Process prove out B - First article inspection C - Conformance lot inspection		
Section 3 Requirement		Section 4 Method	Verification Methods				Verification Class/Qty.		
			1	2	3	4	A	B	C
3.1.1	Process prove out	4.2			X	X	X		
3.1.2	First article inspection	4.3			X	X	-	X	-
3.1.3	Conformance inspection	4.4			X	X	-	-	X
3.2.1	Part coating thickness	4.6.1			X	X	-	1/	1/
3.2.2	Conformity coupon coating thickness	4.6.2			X	X	2 2/	-	-
3.2.3	Coating adhesion	4.6.3				X	2 3/	2 3/	1/
3.2.4	Coating hardness	4.6.4				X	2 3/	1/	1/
3.2.5	Coefficient of friction	4.6.5				X	3 4/	-	-
3.2.6	Resistance to fluids	4.6.6				X	2 3/	-	-
3.2.7	Workmanship	4.6.7			X		100%	100%	1/
3.3.1	Maximum operating temperature	4.7.1				X	5 3/	-	-
3.3.2	Thermal stability	4.7.2				X	2 3/	-	-
3.3.3	Humidity resistance	4.7.3				X	3 3/	-	-
3.3.4	Salt-spray (5 % solution) resistance	4.7.4				X	3 3/	-	-
3.3.5	Water displacement and water stability	4.7.5				X	3 3/	-	-
3.4	Operating requirements	4.8				X	-	See 4.8	See 4.8

Notes:
1/ Refer to applicable drawing.
2/ Sample shall be conformity coupon per 4.5.2.
3/ Sample shall be test panels per 4.5.2.
4/ Sample shall be pins and discs/plates per 4.6.5.

4.1 Classification of inspections. The inspections specified herein are classified as follows:

a. Process prove out (See 4.2).
b. First article inspection (See 4.3).
c. Conformance Inspection (See 4.4).

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4.2 Process prove out. When specified (see 6.2), process prove out shall be performed by examination and tests of all requirements as specified in Table III.

4.2.1 Process prove out rejection. If a sample fails to meet any specified performance requirement, the process prove out shall be rejected.

4.3 First article inspection. When specified (see 6.2), first article inspection shall be performed by examination and tests of all requirements as specified in Table III.

4.3.1 First article inspection rejection. If a sample fails to meet any specified performance requirement, the first article shall be rejected.

4.4 Conformance inspection. Conformance inspection of lot samples shall be accomplished by examinations and tests in accordance with Table III.

4.4.1 Rejection of lots. If a product sample fails to pass any of the specified tests, the production lot represented by that sample, shall be rejected.

4.5 Inspection conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, all examinations and tests shall be performed at a temperature of 25 ± 3°C and at a relative humidity between 30 and 70 percent.

4.5.2 Preparation of test panels, conformity coupons, and pins.

a. Test panels. Test panels. The test panels shall be approximately 0.32 cm by 7.6 cm by 15.2 cm. The test panels shall be steel conforming to AISI 8620 per ASTM A108-18 or ASTM A322-13 with hardness range of HRC 28-40. The test panels shall be carburized to R15N 89.5 to 91.0 with an effective case depth of 0.010” to 0.015”. The test panels shall have a finish of Ra ≤ 1.6 µm. The test panels shall be pre-cleaned and prepared as per vendor coating application procedures.

b. Conformity coupons. Conformity coupons. The conformity coupons shall be in accordance with DWG 13104637. The coupons shall be steel conforming to AISI 8620 per ASTM A108-18 or ASTM A322-13 with hardness range of HRC 28-40. The coupons shall be carburized to R15N 89.5 to 91.0 with an effective case depth of 0.010” to 0.015”. The coupons shall have a finish of Ra ≤ 1.6 µm. The coupons shall be pre-cleaned and prepared as per vendor coating application procedures.

c. Steel pins. The steel cylindrical pins shall be prepared IAW ASTM G99-17 with an end radius of 10 mm. The pins shall be steel conforming to AISI 8620 per ASTM A108-18 or ASTM A322-13 with hardness range of HRC 28-40. The pins shall be carburized to R15N 89.5 to 91.0 with an effective case depth of 0.010” to 0.015” and be provided in the coated and uncoated condition as applicable. The pins shall have a finish of ≤ 1.6 µm. The pins shall be pre-cleaned and prepared per vendor coating application procedure.

d. Aluminum pins. The aluminum cylindrical pins shall be prepared IAW ASTM G99-17 with

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an end radius of 10 mm. The pins shall be anodized 7075 aluminum (as per AMS-QQ-A-367 condition T6 with Protective finish 7.5.2 of MIL-STD-171), with film thickness 0.0010" ±0.0002" and sealed. Grit blast the 7075 aluminum specimens prior to finish to provide a non-reflective matte surface after finish. The aluminum pins shall have a finish of Ra ≤ 3.2 µm.

4.6 Support of ownership verifications.

4.6.1 Part coating thickness. Coating thickness testing shall be conducted in accordance with ASTM D7091-21 or ASTM B499.

4.6.2 Conformity coupon coating thickness. Testing shall be conducted in accordance with ASTM E3-11 as follows: conformity coupons shall be sectioned at three locations. Each section shall be mounted/polished per details 'a' through 'c' below. Coating thickness is measured metallographically using optical or electron microscopy at a minimum of four locations in 90 degree increments around the circumference on the inner diameter and outer diameter to verify thickness and thickness uniformity.

a. Location 1 – 25.4 mm (1 in) from left edge of DWG 13104637
b. Location 2 – 114.3 mm (4.5 in) from left edge of DWG 13104637
c. Location 3 – 139.7 mm (5.5 in) from left edge of DWG 13104637

4.6.3 Coating adhesion. Coating adhesion testing shall be conducted in accordance with ASTM B571-18 method 13.1 with the following modifications: The scribing of the sample is completed with a tungsten carbide tool bit with a 30-degree angle tip using a routing machine or similar. Each scribe is completed in a single pass to penetrate through the coating a minimum of 150% and a maximum of 300% of coating thickness (i.e., 15 µm minimum and 30 µm maximum for 10 µm thick coating). The sample is scribed in a grid pattern with a distance between the centers of the scribed lines of 0.4 mm, with a minimum of 3 cuts per direction. The areas between the scribes within the grid pattern are analyzed at 100X magnification for locations of adhesive failure. If any adhesive failure is detected, the adhesion is considered unacceptable.

4.6.4 Coating hardness. Hardness testing of the coating shall be conducted in accordance with ASTM E2546-15 for Class I and Class II coatings and ASTM E384-17 or ASTM E2546-15 for Class III coatings.

4.6.5 Coefficient of friction. For each panel and pin combination, testing shall be conducted in accordance with ASTM G99-17 pin-on-disk testing with the following modifications:

a. Testing load shall be 5 N with a sliding speed of 0.1 m/s, wear track diameter of 20 mm, conducted for 5000 cycles.
b. A minimum of three tests shall be conducted with the reported value of coefficient of friction determined by the average coefficient of friction from 1000-5000 cycles for each test.
c. The applicable pin shall be in accordance with 4.5.2.
d. The disc/plate specimen shall be comprised of material from test panels. The test panels used shall be in accordance with 4.5.2. The test panels may be cut to any size to facilitate pin-on-disc testing with the following restrictions; (1) Each wear track shall be a minimum of 10 mm from any edge and (2) each track must be separated by a minimum of 10 mm.

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4.6.6 Resistance to fluids. The coating applied to test panels (see 4.5.2) shall be subjected to ASTM D2510-94 Procedure C. Test fluids shall be in accordance with Table 1. When performing ASTM D2510-94, use naphtha, aliphatic followed by acetone or any environmentally safe cleaner that sufficiently cleans surfaces to pass ASTM F22-21. After exposure to fluids, there shall be no visual changes. After exposure to fluids, test in accordance with coating hardness (see 4.6.4).

4.6.7 Workmanship. Coated parts, panels, coupons, pins, plates/discs shall be visually examined to verify workmanship.

4.7 Environmental verifications.

4.7.1 Maximum operating temperature. The coating applied to test panels (see 4.5.2) shall be exposed to 350 ± 5° C in a laboratory oven for a minimum of 1 hour, test in accordance with coating adhesion (4.6.4) and coefficient of friction (see 4.6.5).

4.7.2 Thermal stability. The coating applied to test panels (see 4.5.2) shall be subjected to thermal stability testing conducted in accordance with ASTM D2511-20. After thermal stability, the test panels sample shall be examined visually and microscopically at a magnification of 12X for uniformity in color, smoothness.

4.7.3 Humidity resistance. The coating applied to test panels (see 4.5.2) shall be evaluated for corrosion protection in high humidity in accordance with Method 5329 of FED-STD-791. Corrosion on the outer 6.4 mm of the panels shall not be cause for rejection.

4.7.4 Salt-spray (5 % solution) resistance.

a. Panel edge dressing. The edges of the test panels (see 4.5.2) shall be examined closely, and sanded if necessary, to ensure panel edges are fully rounded so as not to promote corrosion initiating at a sharp edge. No additional protective substance (e.g., wax or asphalt) shall be applied to the panel edges to inhibit corrosion.

b. Salt-spray test. The coating applied to test panels (see 4.5.2) shall be evaluated for salt-spray resistance in accordance with ASTM B117-19 except that the coating test panels (see 4.5.2) shall be placed at an angle of 10- to 20-degrees from the vertical with the four-inch edge with drilled holes facing down in the cabinet during exposure to salt-spray.

c. Salt-spray failure criteria. Visually examine the test panels after exposure for compliance with requirements of 3.3.4. Corrosion on the outer 6.4 mm (1/4 in) of the panels shall not be cause for rejection.

4.7.5 Water displacement and water stability. The coating applied to test panels (see 4.5.2) shall be subjected to the water displacement and water stability test in accordance with Method 3007 of FED-STD-791, except that the test shall be made only on the compound-water mixture.

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4.8 Operating requirement verification.

TABLE IV. Weapon system operating verification.

Weapon System	Verification
M4A1	See Appendix A

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coating covered by this specification is intended for use ferrous metals. It is useful under the following conditions:

a. For sliding motion applications such as plain and spherical bearings, flap tracks, hinges, threads, and cam surfaces.
b. Where conventional lubricants are difficult to apply or retain, or where other lubricants may be easily contaminated with dirt and dust.
c. If mechanisms are operated at infrequent intervals or are lubricated for life.
d. Where long-term corrosion protection is required under static conditions
e. Where a solvent-resistant coating is required.

6.1.1 Corrosion protection life. When the coating is applied over coated steel, under static conditions, the coating can be expected to provide corrosion protection for five years in indoor storage and approximately two years protection in outdoor storage.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of documents referenced (see 2.0).
c. Packaging, packing, and marking data with requirements in detail (see 5.1).
d. Specify application and surface preparation requirements (see appendix).
e. Hazardous material identification and material safety data. The contractor should certify that if any carcinogenic or potentially carcinogenic constituents are present as defined under the Hazard Communication Standards (HCS) 29 CFR 1910.1200, appropriate warnings (see HCS 29

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CFR 1910.1200) should be included on product labels.

f. Ozone depleting substances (ODS) are prohibited from being contained within the coating.
g. When Process prove out is required (see 3.1).
h. When First article inspection is required (see 3.1).
i. Conformance inspection is required (see 3.1).

6.3 Definitions.

6.3.1 Class I weapon stoppage. A failure that is immediately clearable (within 10 seconds or less) by the operator following prescribed immediate action procedures.

6.3.2 Class II weapon stoppage. A failure that is clearable by the operator (requiring more than 10 seconds but not more than 10 minutes) using only the equipment and tools issued with the weapon.

6.3.3 Class III weapon stoppage. A failure of a severe nature that; (1) is operator correctable but requires more than 10 minutes, (2) operator cannot correct and requires assistance (no time limit), (3) requires higher level of maintenance, or authorized operator correction cannot be accomplished because of unavailability of necessary tools, equipment or parts.

6.4 Subject term (key word) listing.

M4A1 Carbine

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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APPENDIX A

M4A1 OPERATING REQUIREMENTS AND VERIFICATIONS.

A.1. SCOPE

A.1.1 Scope. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance. This appendix presents the requirements and verification methods that are unique to M4A1, for use in combination with the rest of the requirements/verifications in the main part of this specification.

A.2. APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in sections A.3 or A.4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections A.3 or A.4 of this specification, whether or not they are listed.

A.2.2 Government documents.

a. Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680 - Degreasing Solvent

(Copies of these documents are available online at https://quicksearch.dla.mil)

A.2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY DEVELOPMENTAL TEST COMMAND

TOP 3-2-045 - Automatic Weapons, Machine Guns, and Hand and Shoulder Weapons

(Copies of these documents are available online at https://www.atec.army.mil/publications/documents.html)

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APPENDIX A

DRAWINGS

USARMY Combat Capabilities Development Command Armament Center

8448507	-	Carrier, Bolt
8443572	-	Computer, Rate of Fire
12972700	-	Carbine, 5.56 MM, M4A1
13020533	-	Cartridge, 5.56MM, Ball, M855A1

(Copies of these drawings may be requested from DEVCOM Armaments Center FCDD-ACE-SP by email at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil)

A.3. REQUIREMENTS

A.3.1 Required inspections

A.3.1.1 Process prove out. When specified (See 6.2), samples shall be subjected to process prove out in accordance A.4.2.

A.3.1.2 First article inspection. When specified (See 6.2), samples shall be subjected to first article inspection in accordance with A.4.3.

A.3.1.3 Conformance inspection. Samples shall be subjected to conformance inspection in accordance with A.4.4.

A.3.2 Coating thickness conformity.

a. For coated M4A1 bolt carrier (DRW 8448507) complex external geometries and internal surfaces with inner diameter L/D ratios of up to 6:1 for diameters as small as 7.6 mm and L/D ratios of up to 15:1 for inner diameters of 25.4 mm, no individual coating thickness reading shall exceed ±25% of the average thickness for Class III coatings (see 1.2.1).

b. For coated M4A1 bolt carrier (DRW 8448507) complex external geometries and internal surfaces with inner diameter L/D ratios of up to 6:1 for diameters as small as 7.6 mm and L/D ratios of up to 15:1 for inner diameters of 25.4 mm, no individual coating thickness reading shall exceed ±50% of the average thickness for Class I and II coatings (see 1.2.1).

A.3.3 Cold temperature. The M4A1s, including coated parts, shall be exposed to severe cold for 18 hours. After exposure, the weapon shall allow no more than zero Class II or III stoppages, shall allow not more than two Class I stoppages in 600 rounds, and shall sustain a rate of fire of not less than 650 rounds per minute (rpm).

A.3.4 Dust environments. The M4A1s, including coated parts, shall be exposed to a dust environment conducted in accordance with provisions for static test procedure for dust exposure of TOP 3-2-045, except as noted. After exposure, the weapon shall allow zero Class II or III stoppages, shall allow not more than six Class I stoppages in 600 rounds, and shall sustain an rate of fire of not less than 600 rpm.

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A.3.5 Salt-spray environments. The M4A1, including coated parts, shall be exposed to a salt-spray environment for 48 hours in accordance with Salt Fog Test Procedure of TOP 3-2-045, except as noted. After exposure, the weapon shall allow zero Class II or III stoppages, shall allow not more than two Class I stoppages in 240 rounds, and shall sustain a rate of fire of not less than 650 rpm. This test is only performed during first article inspection and conformance inspection on the part in the applicable drawing.

A.4. VERIFICATION

TABLE A-I. Requirement/verification cross reference matrix.

METHOD OF VERIFICATION 1 - Analysis 2 - Demonstration 3 - Examination 4 – Test	CLASSES OF VERIFICATION A - Process prove out B - First article inspection C - Conformance lot inspection

Section 3 Requirement		Section 4 Method	Verification Methods				Verification Class/Qty.		
			1	2	3	4	A	B	C
A.3.1.1	Process prove out	A.4.2			X	X	X	-	-
A.3.1.2	First article inspection	A.4.3			X	X	-	X	-
A.3.1.3	Conformance inspection	A.4.4			X	X	-	-	X
A.3.2	Coating thickness conformity	A.4.5.1			X		-	1 1/	1 1/
A.3.3	Cold temperature	A.4.5.2				X	-	3 3/	-
A.3.4	Dust environments	A.4.5.3				X	-	3 3/	-
A.3.5	Salt-spray environments	A.4.5.4				X	-	1 3/	1 3/

Notes:

1/ Sample is one bolt carrier. This test is only required when a coated bolt carrier (DRW 8448507) is being procured.
2/ Refer to applicable drawing.
3/ The rate of fire shall be measured using test equipment in accordance with DRW 8443572.

A.4.1 Classification of inspections. The inspections specified herein are classified as follows:

a. Process prove out (See A.4.2)
b. First article inspection (See A.4.3).
c. Conformance inspection (See A.4.4).

A.4.2 Process prove out. When specified (see 6.2), process prove out shall be performed by analysis, demonstration, examination and tests of all requirements as specified in Table A-I.

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A.4.2.1 Process prove out rejection. If a sample fails to meet any specified performance requirement, the process prove out shall be rejected.

A.4.3 First article inspection. When specified (see 6.2), first article inspection shall be performed by analysis, demonstration, examination and tests of all requirements as specified in Table A-I.

A.4.3.1 First article test rejection. If a sample fails to meet any specified performance requirement, the design shall be rejected.

A.4.4 Conformance inspection. Conformance inspection of lot samples shall be accomplished by examinations, demonstrations and tests in accordance with Table A-I.

A.4.4.1 Rejection of lots. If a product sample fails to pass any of the specified tests, the production lot represented by that sample, shall be rejected.

A.4.5 Verifications.

a. Weapon performance. The performance of the candidate coating shall be tested by live-fire weapons tests in accordance with TOP 3-2-045, except as noted herein.

b. Test weapons. The test weapon shall be the CARBINE, 5.56 MILLIMETER: M4A1 DRW (12972700). New weapons (guns that have been fired less than 5000 rounds) shall be used.

c. Inspection and preparation of test weapons.

i. Inspection. Disassemble the M4A1 and inspect the headspace, bore erosion, chamber dimensions, firing pin protrusion and indentation, trigger pull and overall condition.

ii. Cleaning and preparation. While disassembled, clean all traces of lubricant and dirt by washing with a volatile mineral spirit, such as MIL-PRF-680, Degreasing Solvent.

d. Validation of test weapons. Each weapon shall then test fire 120 rounds in 3-5 round bursts, in an ambient temperature from 10 to 32 °C (50 to 90 °F). Any weapon that fails to achieve an average baseline cyclic rate of 700 to 970 rpm shall be disqualified for use as a test weapon.

e. Ammunition. Ammunition shall be M855A1 (DRW 13020533) loaded in 30 round magazines. The ammunition required for each of the tests shall remain outside the individual environmental chambers, and shall not be conditioned.

f. Firing test summary. Summary weapons firing data is presented in TABLE A-2.

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TABLE A-2. Weapons firing test summary.

Firing Test	Application (lubricant)	Test Condition	Total Rds / Weapon	Failure criteria 1/
Inspection	None	Ambient	120	A firing rate no less than 700 rpm and not greater than 970 rpm
Cold temp (5 cycles)	None	-51 ± 2 ºC (-60 ± 4 ºF)	600	No more than two Class I stoppages per weapon and a firing rate no less than 650 rpm
Dust (5 cycles)	None	After dusting Ambient temp	600	No more than six Class I stoppages per weapon and a firing rate no less than 600 rpm
Salt-spray (2 cycles) (48/96hrs)	None	After exposure Ambient temp	240	No Class I stoppages per weapon and a firing rate no less than 650 rpm
1/ Note: Any Class II or III weapon stoppage during any test is cause for failure of the test.				

A.4.5.1 Coating thickness conformity. The coated M4A1 bolt carrier shall be sectioned at three locations. Each section shall be mounted/polished per details ‘a’ through ‘c’ below. Coating thickness is measured at a minimum of four locations in 90 degree increments around the circumference on the inner diameter and outer diameter to verify thickness and thickness uniformity. This test is only performed during first article inspection and conformance inspection.

a. Location 1 – 25.4 mm from the forward face of the bolt carrier (DRW 8448507).
b. Location 2 – 114.3 mm from the forward face of the bolt carrier (DRW 8448507).
c. Location 3 – 139.7 mm from the forward face of the bolt carrier (DRW 8448507).

A.4.5.2 Cold temperature. Three M4A1s, including coated parts, shall be used for cold temperature testing. This test is only performed during first article inspection.

a. Inspection and preparation. The test weapons shall be disassembled, cleaned, and then reassembled. After reassembling the weapons (with bolts and ejector ports closed), the weapons shall be conditioned at -51 ± 2 °C (-60 ± 4 °F) for 16 ± 1 hours.

b. Weapons firing. Weapons firing shall be conducted within the cold temperature chamber immediately after the first conditioning cycle is complete. The weapon shall be loaded with an un-conditioned (ambient temperature) ammunition. Each magazine shall be fired in 3-5 round bursts to check for stoppages. Each cycle shall consist of firing 4 magazines. The 4th magazine shall be fired in 3-5 round bursts to validate the firing rate. Repeat the cycle, except condition the weapon for only 2 hours ± 15 minutes at the same temperature. Fire the remaining rounds in the same manner employed in the first cold temperature firing cycle, for a total 600 rounds fired from each weapon.

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c. Cold temperature failure criteria. Any Class II or III weapon stoppage, more than two Class I stoppages in 600 rounds, or a reduction in the rate of fire below 650 rpm, shall be cause for rejection of the coating. Individually, each weapon shall meet the stoppages and rate of fire requirement.

A.4.5.3 Dust environments. Three M4A1s, including coated parts, shall be used for dust testing. This test shall be conducted in accordance with provisions for static test procedure for dust exposure of TOP 3-2-045, except as noted. This test is only performed during first article inspection.

a. Inspection and preparation. The test weapons shall be disassembled, cleaned, and then reassembled. After reassembling the weapons (with bolts and ejector ports closed, and barrel dust caps installed, or muzzles taped), the weapons shall be placed in weapons racks in a vertical orientation in the chamber, and exposed to blowing dust. Dusting shall be as specified in TOP 3-2-045, except that the dusting cycle time shall be 10 minutes.

b. Weapons firing. After the first dusting cycle, the weapon shall be moved to the firing position, loaded with an un-conditioned 120 rounds and fired. Each magazine shall be fired in 3-5 round bursts to check for stoppages. Each cycle shall consist of firing 4 magazines. The 4th magazine shall be fired in 3-5 round bursts to validate the firing rate. This conditioning and firing sequence (no disassembly or cleaning) shall be repeated 4 times for a total of 50 minutes of dust exposure, and 600 rounds fired from each weapon.

c. Dust test failure criteria. Any Class II or III weapon stoppage, more than six Class I stoppages in 600 rounds, or a reduction in the rate of fire below 600 rpm, shall be cause for rejection of the coating. Only the two weapons with the least number of stoppages shall be used to determine if the requirement is met. Individually, each of the two down selected weapons shall meet the stoppages and rate of fire requirement.

A.4.5.4 Salt-spray environment test. One M4A1, including coated parts, shall be subjected to salt-spray testing and its performance alone shall determine if the requirement is met. This test is only performed during first article inspection and conformance inspection.

a. Inspection and preparation. The test weapon shall be disassembled, cleaned, and then reassembled. After reassembling the weapon (with bolts and ejector ports closed), the weapon shall then be placed in a salt-spray cabinet and exposed to 5 % salt-spray in accordance with the Salt Fog Test Procedure of TOP 3-2-045. Testing shall consist of two alternating 24-hour exposure 24-hour drying periods, for a total of 48 hours of salt-spray exposure and 48 hours of drying, 96 hours in total.

b. Weapon firing. After each 48 hour cycle, the weapon shall be moved to the firing position and loaded with 120 un-conditioned rounds and fired. Each magazine shall be fired in 3-5 round bursts to check for stoppages. Each cycle shall consist of firing 4 magazines. The 4th magazine shall be fired in 3-5 round bursts to validate the firing rate. This conditioning and firing sequence shall be repeated for a total of 240 rounds fired from each weapon.

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c. Salt-spray environment failure criteria. Any Class II or III weapon stoppage, more than zero Class I stoppages in 240 rounds, or a reduction in the rate of fire below 650 rpm, shall be cause for rejection of the coating. The weapon shall meet the stoppages and rate of fire requirement.

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - AS
Air Force - 11

Preparing activity:
Army - AR

(Project MFFP-2023-007)

Review activities:
Army - AR, AV, MI, SM, MD
Navy - SH, MC
Air Force - 68
DLA - GS, PS

Industry associations:

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